ARCUS
INFRASTRUCTURE INTELLIGENCE

PATH 01 - NEW TERRITORY

Territory Briefing: Torino

Professional province-level preliminary briefing: historical evidence, Atlas map, project-context focus and operational next checks.

INFRASTRUCTURE INTELLIGENCE / GENERATED 09/06/2026

REPORT ID	VERSION	ANALYSIS	SPATIAL LEVEL	GENERATED BY
ARCUS-P01-20260609-torino	ARCUS Professional v1.0-preview	Territory Briefing	Province level	ARCUS Professional

01 EXECUTIVE SUMMARY

ARCUS is based on the first systematic database of collapsed bridges in Italy from 2000 to today. The value is not map aggregation: it is the translation of documented collapse evidence into operational infrastructure intelligence.

ARCUS identifies 42 documented cases in the selected province and translates the Step 3 exposure indicators into operational actions for Bridge interventions. The PDF map shows ARCUS cases; the hazard-layer contribution is summarised through the indicators below. The dominant territorial signal is **Hydraulic**; the leading historical driver is **Hydraulic**.

Recommended use: province-level preliminary screening for Bridge interventions, before design, due diligence or site-specific investigation.

> The value is not the case list: it is understanding whether Torino has already shown concentrated vulnerability under extreme conditions or risk distributed over time.

FINDING 1

Asset: first systematic database of collapsed bridges in Italy from 2000 to today.

FINDING 2

The hydraulic context dominates the territorial reading.

FINDING 3

Collapse Rate: 5.82x the national ARCUS/AINOP rate, separate from Priority Index.

FINDING 4

41 triggered events and 30 total collapses shape the technical priority.

FINDING 5

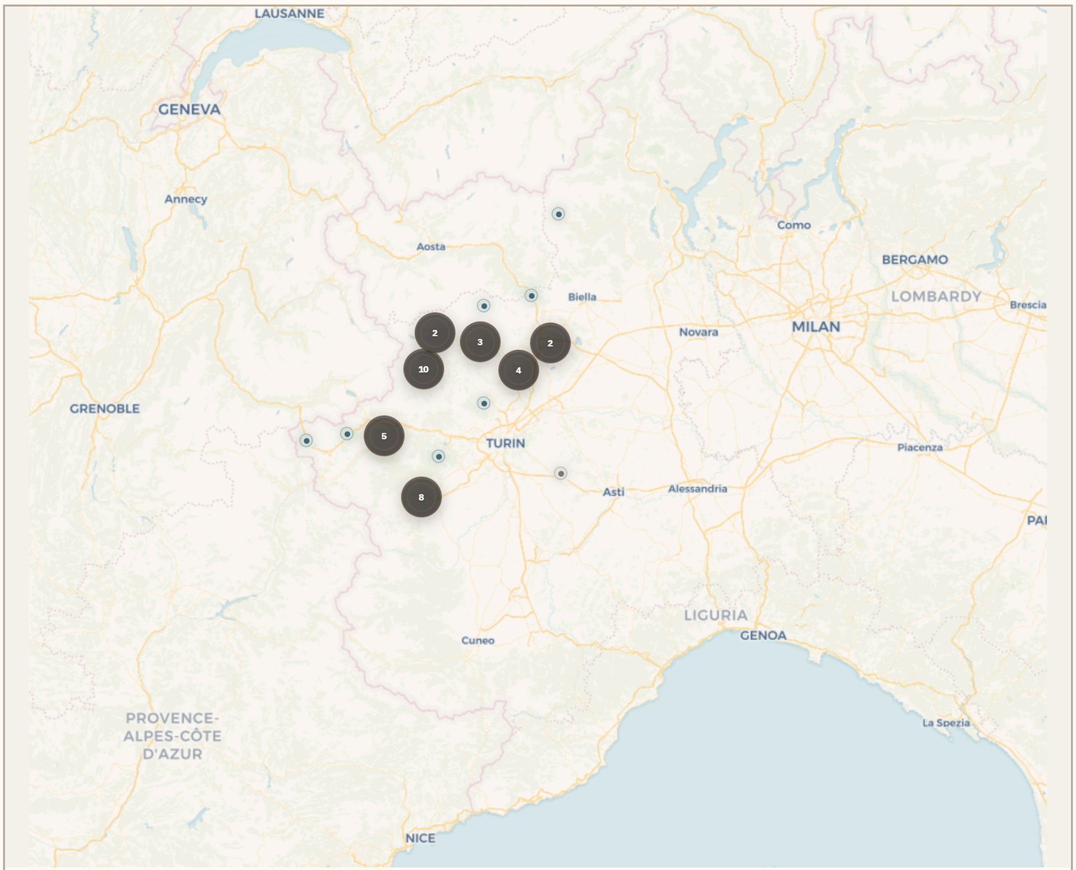
Very high evidence reliability with 125 linked sources.

WHY ARCUS FLAGS THIS

- ARCUS is based on the first systematic database of collapsed bridges in Italy from 2000 to today. The value is not map aggregation: it is the translation of documented collapse evidence into operational infrastructure intelligence.
- Dominant exposure: Hydraulic (97 / 100).

- Historical severity: 30 total collapses and 41 triggered events.
- Map/appendix cases: P1 Castellamonte, P2 Ala di Stura, P3 Ala di Stura.
- Evidence window: 2000-2024 (25 years of evidence).
- Evidence package: 125 linked sources supporting the signal.

02 ARCUS CASES MAP



The PDF map shows ARCUS georeferenced cases within the selected province. Hazard overlays are summarised through the exposure indicators below. Full interactive layer control is available in ARCUS Professional Atlas.

Static ARCUS cases map - full interactive view available in Professional Atlas.

03 HOW TO READ

ARCUS does not produce a generic score: it reads what happened to Italian bridges from 2000 to today, why it happened, and what to check before deciding.

> The report uses language focused on what to check before design. The Priority Index appears at the end: it is the conclusion of the reading, not the starting point.

REPORT CONFIGURATION

New construction

The report explains what to check before design.

EVIDENCE BASE

42 ARCUS events

ARCUS is based on the first systematic database of collapsed bridges in Italy from 2000 to today. The value is not map aggregation: it is the translation of documented collapse evidence into operational infrastructure intelligence.

04 BLOCK 1 / TERRITORY

Torino combines alpine and valley settings with torrential watercourses and lowland areas crossed by major rivers. The dominant signal should therefore be read as differentiated hydraulic exposure: rapid floods and sediment/debris transport in mountain basins, overflow dynamics and major crossings in the plain.

> WMS layers remain separate exposure indicators. At this stage they are not merged into a single number.

Hydraulic exposure	**97**
Structural Vulnerability Exposure	**30**
Landslide exposure	**28**
Seismic exposure	**28**

05 **BLOCK 2 / HERITAGE**

HERITAGE VULNERABILITY

Collapse Rate

5.82x

3.268 ARCUS cases per 100 AINOP bridges; denominator 1285 counted bridges.

CONFIDENCE: HIGH

AINOP COVERAGE

high

1285

AINOP denominator is broad enough for provincial relative benchmarking.

This places Torino among the provinces with the highest documented collapse rate in Italy (national rank 12).

ARCUS/AINOP Collapse Rate: 3.268 ARCUS cases per 100 AINOP bridges; 5.82x the national ARCUS/AINOP rate. Denominator: 1285 AINOP bridges (1285 road, 0 rail).

06 **BLOCK 3 / HISTORICAL PATTERN**

DOMINANT TRIGGER	TEMPORAL DISTRIBUTION	GEOMORPHOLOGY	RELIABILITY
98%	**Concentrated flood scenario**	**Hydraulic**	**96 / 100**
HYDRAULIC	2000-2024	HYDRAULIC	VERY HIGH
share of the provincial ARCUS sample	collapses concentrated in a single flood/extreme-hydraulic scenario (2000-10-13, 88%).	mainly tied to watercourses, flood dynamics, scour and sediment/debris transport.	125 linked sources

> Note: the cases selected for map/appendix refer to the same 13-16 October 2000 flood event in Piemonte and Valle d'Aosta. This should be read as documented extreme-flood vulnerability, not independent recurrent events.

> Individual cases remain available on the map and in the appendix: the main report presents the pattern, not a case ranking.

ARCUS READING

ARCUS reading: for new Bridge interventions, the Hydraulic signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions in the province of Torino. The report uses language focused on what to check before design.

ATTENTION POINTS

- PRELIMINARY HYDRAULIC COMPATIBILITY
- SCOUR SUSCEPTIBILITY
- FOUNDATION, PIER AND ABUTMENT EXPOSURE
- RIVERBED DYNAMICS AND SEDIMENT/DEBRIS TRANSPORT
- FLOOD/DEBRIS SCENARIOS AND INSPECTION ACCESSIBILITY

DATA REQUESTS

Hydraulic model or preliminary hydraulic study; Riverbed survey / bathymetry where relevant; Geotechnical investigation and foundation concept

ACTION 1

Use the Step 3 exposure indicators to identify the dominant Hydraulic signal.

ACTION 2

Review the documented concentrated historical scenario and use the map/appendix only as case reference.

ACTION 3

Request the minimum technical data package.

ACTION 4

Commission site-specific hydraulic and geotechnical checks.

DATA 1

Hydraulic model or preliminary hydraulic study

DATA 2

Riverbed survey / bathymetry where relevant

DATA 3

Geotechnical investigation and foundation concept

DATA 4

Historical flood levels

DATA 5

Inspection accessibility constraints

1. Use the Step 3 exposure indicators to identify the dominant Hydraulic signal.
2. Review the documented concentrated historical scenario and use the map/appendix only as case reference.

3. Request the minimum technical data package.
4. Commission site-specific hydraulic and geotechnical checks.
5. Use CSV and GeoJSON exports for technical coordination.

08 FINAL PRIORITY INDEX

PRIORITY INDEX

87 / 100

HIGH ATTENTION

Final synthesis after territory, heritage, pattern and actions.

TERRITORIAL EXPOSURE

82 / 100

HYDRAULIC

Hydraulic, landslide and seismic layers read separately.

COLLAPSE RATE

5.82x

CONFIDENCE: HIGH

ARCUS/AINOP bridge-stock signal.

HISTORICAL PATTERN

Concentrated flood scenario

HYDRAULIC

collapses concentrated in a single flood/extreme-hydraulic scenario (2000-10-13, 88%).

> The Priority Index aggregates hydraulic, landslide, seismic exposure and AINOP Collapse Rate. It should be read as the methodological conclusion, not as a substitute for the four blocks above.

09 DATA COVERAGE & LIMITATIONS

> Province-level screening: not design scale.

> ARCUS does not certify structural safety or asset condition.

> Public overlays guide priorities and must be followed by site-specific checks.

> ARCUS currently reports the provincial evidence window as 2000-2024 (25 years of evidence). The Collapse Rate is a derived ARCUS/AINOP indicator: AINOP is fed by owners/managers and may under-represent local assets, especially in small mountain municipalities. For this province the denominator confidence is high: AINOP denominator is broad enough for provincial relative benchmarking.

10 ARCUS CASE APPENDIX

Preview of the most relevant ARCUS cases. The map and CSV/GeoJSON exports remain the correct level for individual case review.

P1	**B00.10.15 - Castellamonte** 2000-10-13 / Hydraulic	Total Collapse / A
P2	**B00.10.03 - Ala di Stura** 2000-10-13 / Hydraulic	Total Collapse / A
P3	**B00.10.05 - Ala di Stura** 2000-10-13 / Hydraulic	Total Collapse / A

A SOURCE APPENDIX

Main sources linked to events in the selected province, deduplicated by title or URL. Total linked sources in this briefing: 125. The full source table can be exported separately from ARCUS Professional.

B00.10.15, B00.10.03, B00.10.05 +34 MORE / ARPA PIEMONTE
Evento alluvionale regionale 13-16 ottobre 2000 – monografia ufficiale
arpa.piemonte.it / PDF

B00.10.15, B00.10.03, B00.10.05 +34 MORE / ANSA
10 anni fa alluvione Nord-Ovest. fu inferno d'acqua – Piemonte e Valle d'Aosta
ansa.it / HTML

B00.10.15, B00.10.03, B00.10.05 +37 MORE / JOURNAL PAPER
Bridge collapses in Italy across the 21st century: survey and statistical analysis
doi.org

B05.11.01 / CORRIERE DELLA SERA
Bloccata la linea ferroviaria Torino-Roma
corriere.it / SHTML

B05.11.01 / GAZZETTA D'ASTI
Crollo della passerella a Villanova: condanne in Tribunale
gazzettadasti.it

B08.05.02 / REGIONE PIEMONTE
Evento alluvionale 29-30 maggio 2008 – relazione tecnica ufficiale
regione.piemonte.it / PDF

B08.05.02 / METEONETWORK
Alluvione maggio 2008 in Provincia di Cuneo: resoconto completo
meteonetwork.it / PDF

B17.06.01 / TORINOTODAY
Cedimento strutturale: è crollato lo storico ponte di Strambino
torinotoday.it / HTML

15 METHODOLOGY SNAPSHOT

ARCUS combines historical collapse evidence, source reliability, spatial relevance, severity, triggers, cause similarity, territorial hazard context and public WMS overlays. The briefing supports preliminary screening and prioritisation: it is not a structural safety certification.

COMPONENT	USE IN THE BRIEFING
Historical evidence	Documented ARCUS records, causes, triggers and severity.
Source reliability	Number, role and quality of available sources.
Spatial relevance	Selected province and administrative provincial boundary.
Hazard context	Public hydraulic/landslide WMS overlays and seismic context.
ARCUS historical pattern	Reading of dominant trigger, temporal concentration, involved geomorphology and source quality. Individual cases remain references in the map, appendix and exports.

16 SCORE AND CLASS LEGEND

CLASS	INTERPRETATION
A	High reliability: multiple sources, preferably institutional or technical.
B	Medium reliability: documented event with limited coverage or source variety.
C	Limited reliability: local, incomplete or to-be-strengthened evidence.
Critical	Maximum attention: strong historical and/or territorial evidence.
High	High attention: significant evidence but not necessarily complete.
Medium / Low	Moderate or low attention, to be read together with available data.